

Ecosystems

Living things interact with each other to form communities called ecosystems. Each ecosystem has a particular character formed by factors such as the climate, geology, and the species that live in it. As plants grow, animals eat the plants, and other animals eat those animals, energy is passed from species to species in a food chain.

ENERGY AND NUTRIENTS

Every ecosystem needs energy and a supply of the nutrients that living things use to build tissue. Most of the energy comes from sunlight, which trees and microbes in this forest soak up and use to fuel their growth. The nutrients include water, carbon, and nitrates (nitrogen and oxygen), along with other chemical compounds that have been dissolved from the rocks by water in the soil or sea.



PRODUCERS AND CONSUMERS

Green plants and many microbes are food producers. They use photosynthesis to turn carbon dioxide and water into carbohydrates such as sugar. They use other raw nutrients to make proteins. Animals eat plants and other living things, and use the digested carbohydrates and proteins to build their own bodies—so they are consumers. The interaction between producers and consumers is the basis of all ecosystems.

▼ FOOD FACTORY

This caterpillar is consuming a leaf that the plant has made from raw chemicals.